

SLOPE STABILITY ANALYSIS BY FINITE ELEMENTS

A guide to the use of Program **slope64**

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Running the **slope64** program

The source code for **slope64** is written in Fortran 2003, but users are provided with the executable file **slope64.exe** and the example data files **ex1.dat** thru **ex7.dat** as described in this report.

The files needed to run the slope program with one of the examples are as follows:

slope64.exe	An executable file of the slope64 program
ex?.dat	A typical example data file as described in this report

Running an example problem

In order to run **slope64** with an example data file, e.g. **ex4.dat**, double-click on the **slope64.exe** icon in Windows explorer which should open up a prompt that States “**Please enter the base name of data file:**”

Then type the base name of the data file **ex4** and press **enter**

If the program runs properly, you should see evidence of iterations being reported back to the screen. When the job is complete you should see the following four additional files in the same folder:

The ex4.res	file is simple text reporting the numerical results of your analysis
The ex4.msh	file is a PostScript file showing the finite element mesh
The ex4.dis	file is a PostScript file showing the deformed mesh at failure
The ex4.vec	file is a PostScript file showing the nodal displacement vectors at failure

Running your own problem

Create your own data file. This report explains the layout of data for use with **slope64**. It may help to make a copy of one of the example data files closest to your own problem and edit that as needed. Let us assume your data file is called **fred.dat**. Note that the extension must be of type **.dat**.

In order to run your own data file **fred.dat**, once more double- click on the executable icon **slope64.exe**, and when prompted type the basename of the data file, namely **fred**

If all goes well, the following additional files will appear in your folder: **fred.res**, **fred.msh**, **fred.vec** and **fred.dat**

Note: Program **slope64.f95** is based closely on **p64.f95** in the textbook, “Programming the Finite Element Method” by I.M. Smith, D.V. Griffiths and L. Margetts. 5th ed., 2014. Users of **slope64** are encouraged to refer to this text and the companion paper, “Slope stability analysis by finite elements”, by D.V. Griffiths and P.A. Lane, *Géotechnique* 49, no.3, pp.387-403, (1999).

Explanation of data for Program **slope64.exe**

A typical configuration is shown in Figure 1.

Slope geometry data:

w1 = Width of top of embankment
s1 = Width of sloping portion of embankment
w2 = Distance foundation extends to right of embankment toe
h1 = Height of embankment
h2 = Thickness of foundation layer

Element discretization data:

nx1 = Number of x-elements in embankment
nx2 = Number of x-elements to right of embankment toe
ny1 = Number of y-elements in embankment
ny2 = Number of y-elements in foundation

Soil property data: ¹

np_types = Number of different property groups
phi,c,psi,gamma,e,v = Material properties $\phi', c', \psi, \gamma, E, \nu$ (**np_types** times)
etype = Property group assigned to each element (**nels** times)
(data not needed if **np_types=1**)

Pseudo-static analysis: ²

k.h = Horizontal acceleration factor

Free surface data: ³

nosurf = Number of free surface coordinates
x,y = x - and y - coordinates of free-surface (**nosurf** times), or the r_u value
gam_w = Unit weight of water, γ_w

Iteration ceiling:

limit = Iteration ceiling (suggested value, 500)

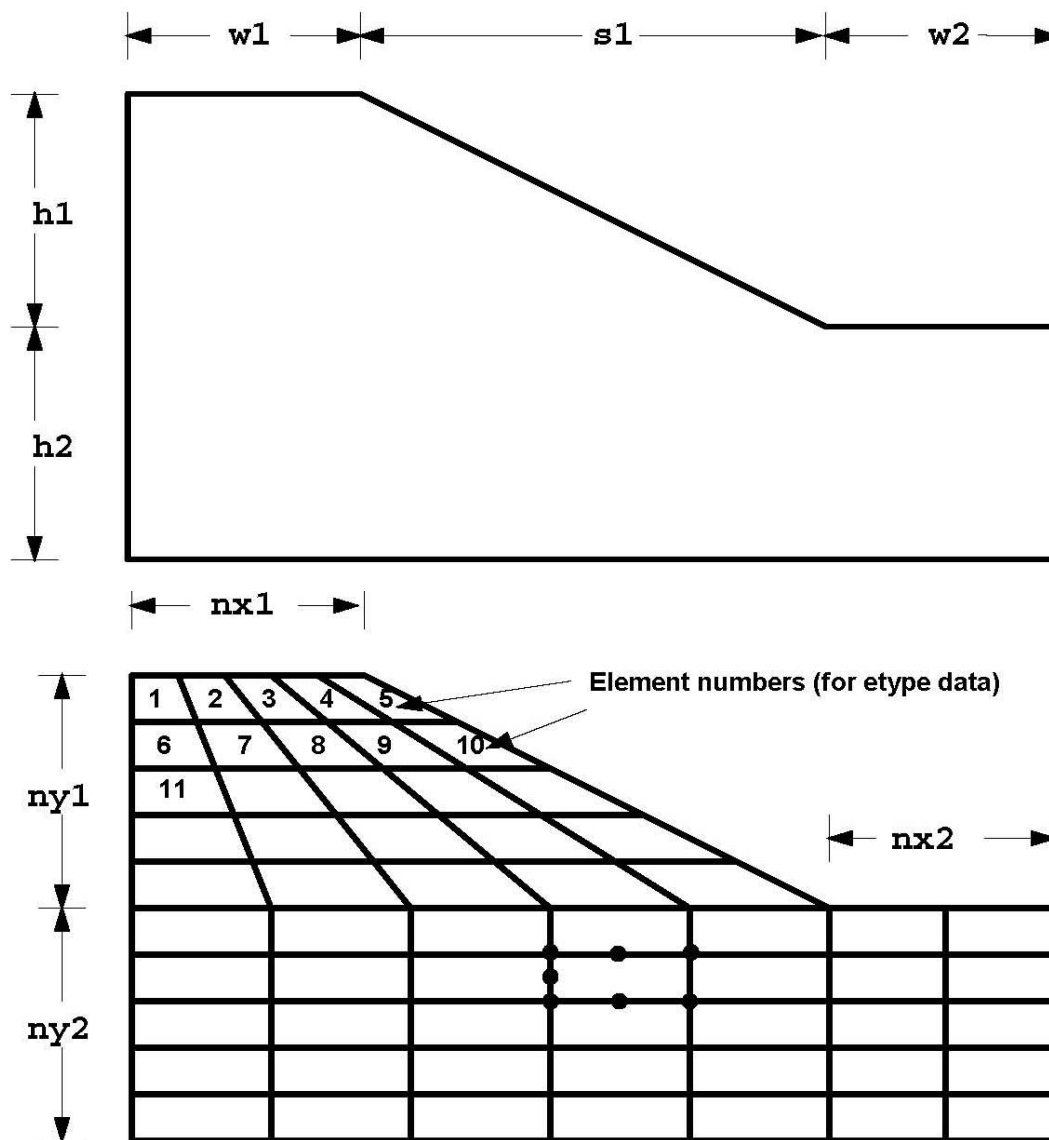
Factor of Safety Tolerance:

fos_tol = Factor of safety tolerance (suggested value, 0.02)

¹ ϕ' is the effective friction angle; c' is the effective cohesion; ψ is the dilation angle and can usually be set to zero; γ is the total unit weight; E is Young's modulus and is often set to a nominal value (e.g. 10^5); ν is Poisson's ratio and is often set to a nominal value (e.g. 0.3), **nels** is the total number of elements in the mesh and is computed internally by the program

² k_h is the horizontal pseudo-static acceleration factor, e.g. for a horizontal acceleration of $0.2g$, set $k_h = 0.2$

³If **nosurf=1**, then instead of **x,y** data, read a single value of r_u .



Number of elements in each section

Fig. 1 Layout and dimensions of embankment geometry

Example 1: A homogeneous slope

The stability analysis is of the homogeneous c' - ϕ' slope shown in Figure 1.1.

Data for Example 1 (ex1.dat)

```
"Example 1: A homogeneous slope"

"Width of top of embankment (w1)"
12.0

"Width of sloping portion of embankment (s1)"
20.0

"Distance foundation extends to right of
embankment toe (w2)"
12.0

"Height of embankment (h1)"
10.0

"Thickness of foundation layer (h2)"
10.0

"Number of x-elements in embankment (nx1)"
32

"Number of x-elements to right of embankment toe (nx2)"
12

"Number of y-elements in embankment (ny1)"
10

"Number of y-elements in foundation (ny2)"
10
```

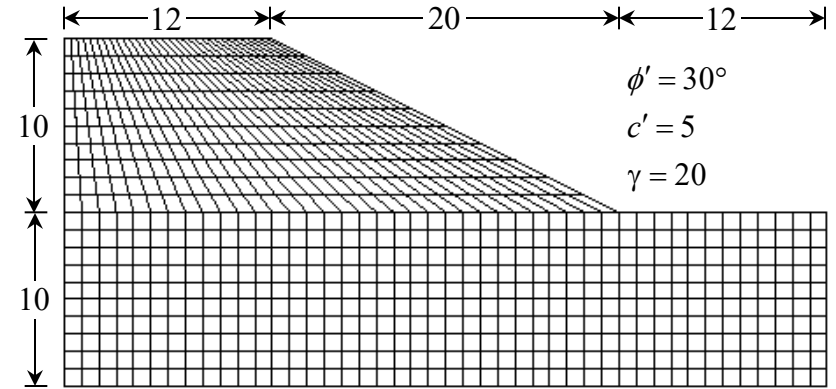


Fig 1.1 Finite element mesh for Example 1

"Number of different property groups (np_types)"

1

"Material properties (phi,c,psi,gamma,e,v) for each group"

30.0 5.0 0.0 20.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.0

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

0

"Unit weight of water (gam_w)"

0.0

"Iteration ceiling (limit)"

1000

"Factor of Safety accuracy tolerance (fos_tol)"

0.02

Output for Example 1 (ex1.res)

trial factor	max displ	iterations
0.5000	0.3050E-01	2
1.0000	0.3050E-01	7
1.5000	0.3546E-01	45
1.5312	0.3622E-01	305
1.5469	0.3651E-01	485
1.5625	0.4026E-01	1000

Estimated Factor of Safety = 1.56

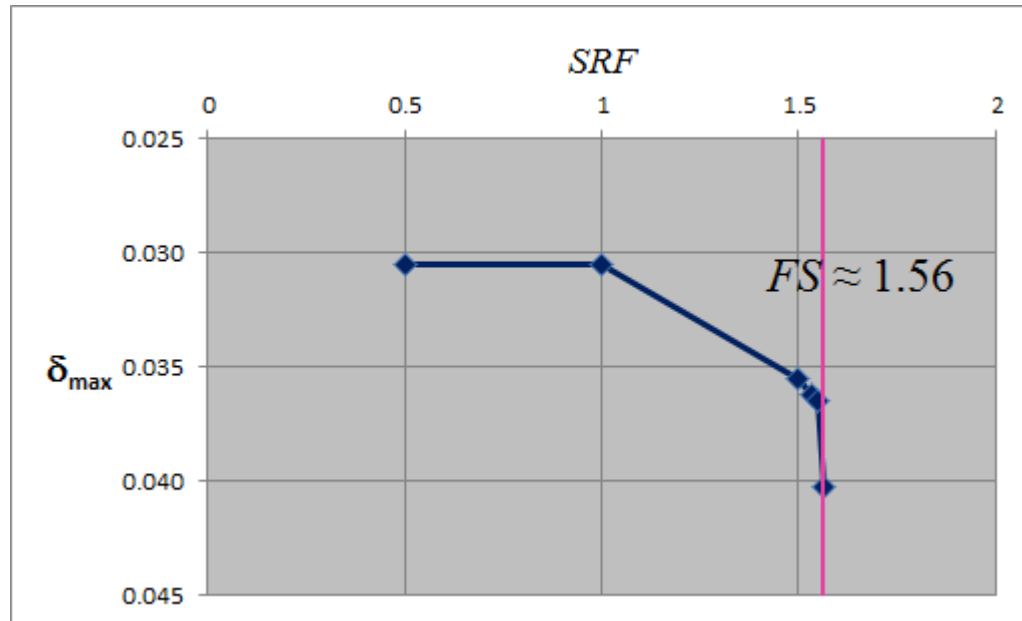


Fig 1.2 SRF vs. $\delta_{\max}$ for Example 1

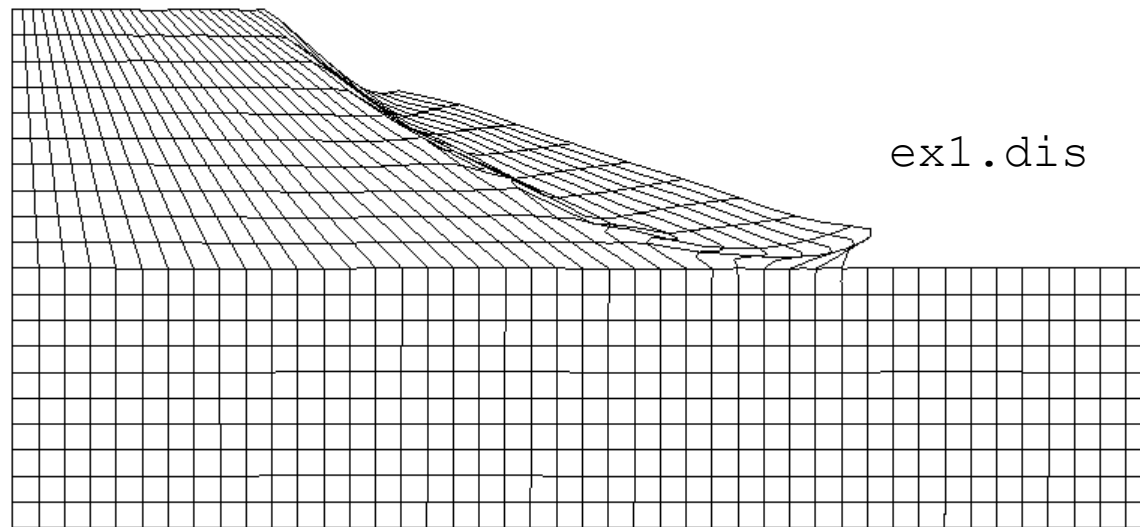
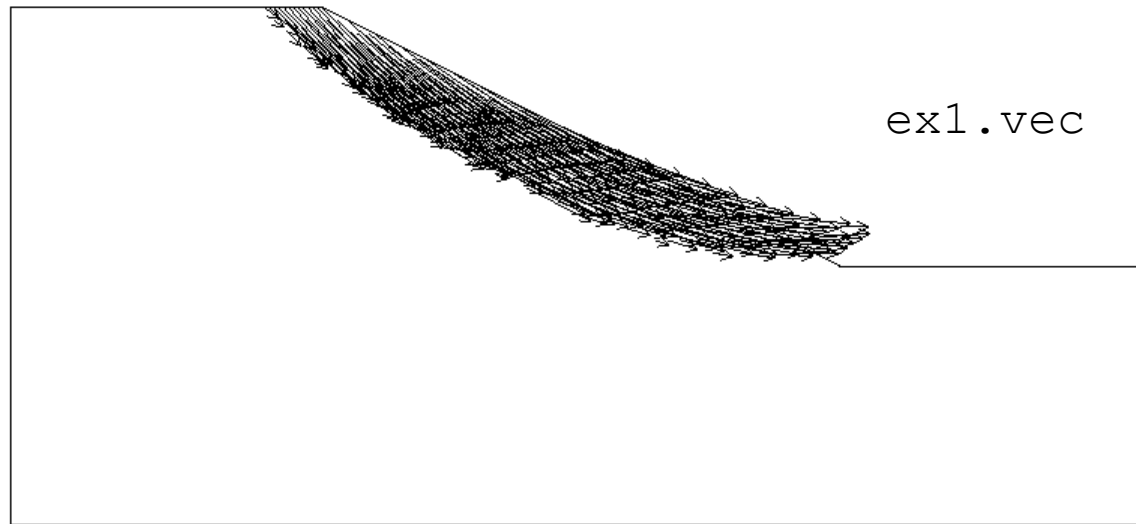


Fig 1.2 Displacement vectors and deformed mesh at failure for Example 1

Example 2: A two-layer slope

The stability analysis is of a two-layer $c'-\phi'$ slope consisting of a stronger soil in the embankment overlying a weaker soil in the foundation as shown in Figure 2.1.

Data for Example 2 (ex2.dat)

"Example 2: A two-layer slope"

"Width of top of embankment (w1)"

1.2

"Width of sloping portion of embankment (s1)"

2.0

"Distance foundation extends to right of embankment toe (w2)"

1.2

"Height of embankment (h1)"

1.0

"Thickness of foundation layer (h2)"

1.0

"Number of x-elements in embankment (nx1)"

5

"Number of x-elements to right of embankment toe (nx2)"

5

"Number of y-elements in embankment (ny1)"

5

"Number of y-elements in foundation (ny2)"

5

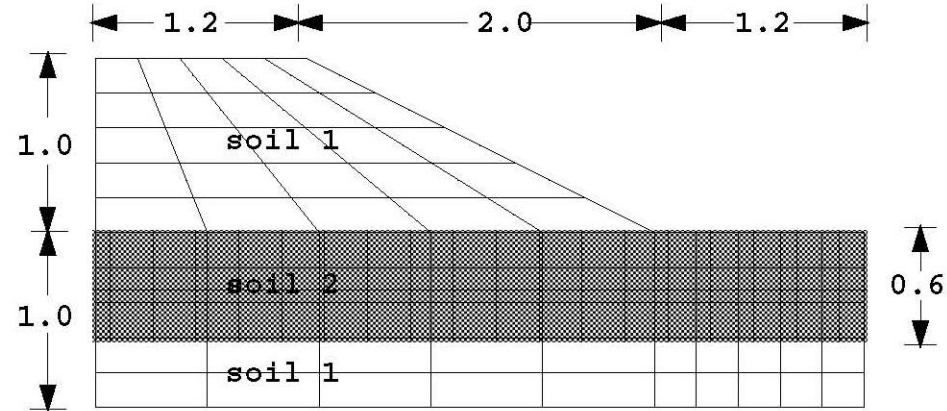


Fig 2.1 Finite element mesh for Example 2

Two-layer

Soil 1

$$\phi' = 25^\circ$$

$$c' = 1.0$$

$$\gamma = 20.0$$

Soil 2

$$\phi' = 15^\circ$$

$$c' = 0.5$$

$$\gamma = 20.0$$

"Number of different property groups (np_types)"

2

"Material properties (phi,c,psi,gamma,e,v) for each group"

25.0 1.0 0.0 20.0 1.e5 0.3

15.0 0.5 0.0 20.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

2 2 2 2 2 2 2 2 2 2

2 2 2 2 2 2 2 2 2 2

2 2 2 2 2 2 2 2 2 2

1 1 1 1 1 1 1 1 1 1

1 1 1 1 1 1 1 1 1 1

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.0

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

0

"Unit weight of water (gam_w)"

9.81

"Iteration ceiling (limit)"

500

"Factor of safety tolerance (fos_tol)"

0.05

Output for Example 2 (ex2.res)

trial factor	max displ	iterations
0.5000	0.3050E-03	2
1.0000	0.3507E-03	33
1.1250	0.3731E-03	51
1.1875	0.3954E-03	199
1.2031	0.4150E-03	348
1.2188	0.4573E-03	500

Estimated Factor of Safety = 1.22

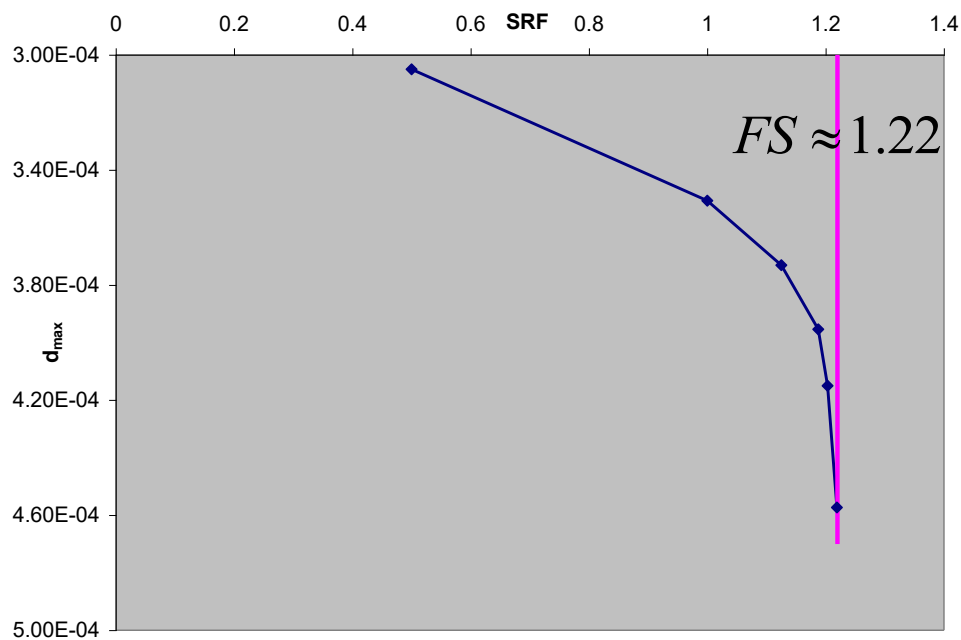


Fig 2.2 SRF vs. $d_{\max}$ for Example 2

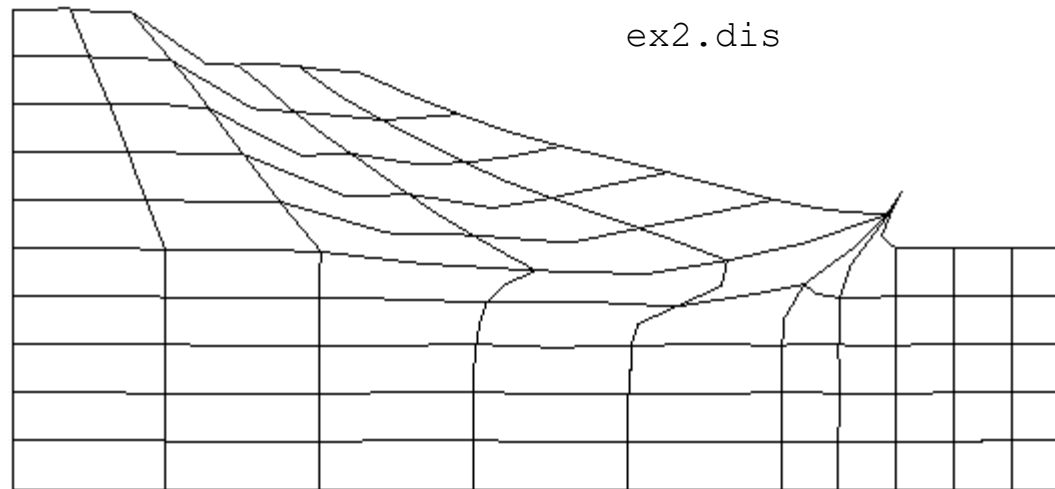
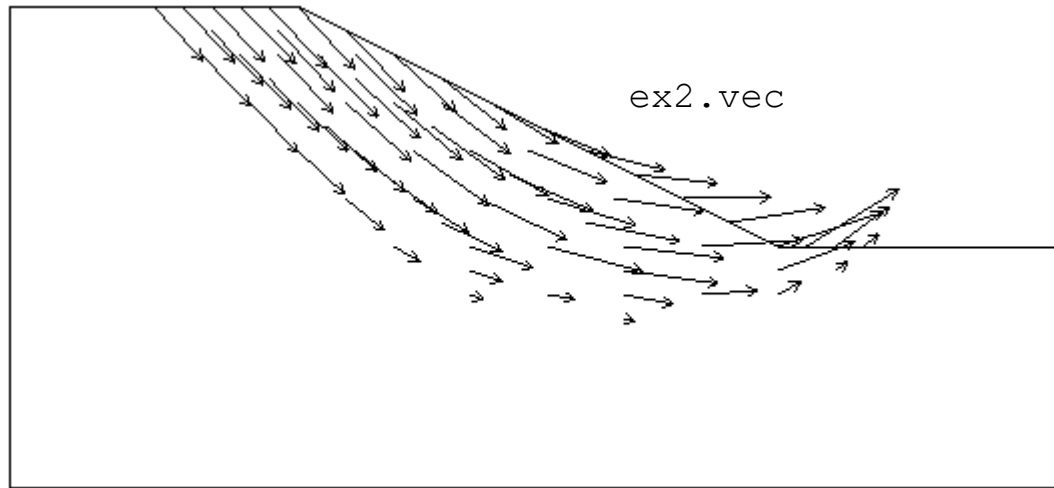


Fig 2.3 Displacement vectors and deformed mesh at failure for Example 2

Example 3: A two-layer undrained clay slope

The stability analysis is of a two-layer undrained clay slope consisting of embankment overlying a stronger soil in the foundation as shown in Figure

Data for Example 3 (ex3.dat)

"Example 3: A two-layer undrained clay slope"

"Width of top of embankment (w1)"

20.0

"Width of sloping portion of embankment (s1)"

20.0

"Distance foundation extends to right of embankment toe (w2)"

20.0

"Height of embankment (h1)"

10.0

"Thickness of foundation layer (h2)"

10.0

"Number of x-elements in embankment (nx1)"

20

"Number of x-elements to right of embankment toe (nx2)"

10

"Number of y-elements in embankment (ny1)"

5

"Number of y-elements in foundation (ny2)"

5

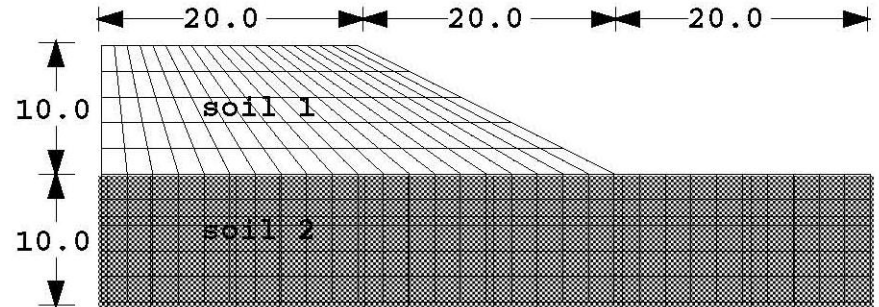


Fig 3.1 Finite element mesh for Example 3

Two-layer Undrained Clay

Soil 1

$\phi_u = 0$

$c_u = 50.0$

$\gamma = 20.0$

Soil 2

$\phi_u = 0$

$c_u = 73.1$

$\gamma = 20.0$

Output for Example 3 (ex3.res)

trial factor	max displ	iterations
0.5000	0.3044E-01	2
1.0000	0.3535E-01	6
1.5000	0.4629E-01	35
1.7500	0.5365E-01	67
1.8750	0.5784E-01	82
1.9375	0.6063E-01	107
1.9688	0.6244E-01	143
1.9844	0.6389E-01	195
2.0000	0.7257E-01	500

Estimated Factor of Safety = 2.00

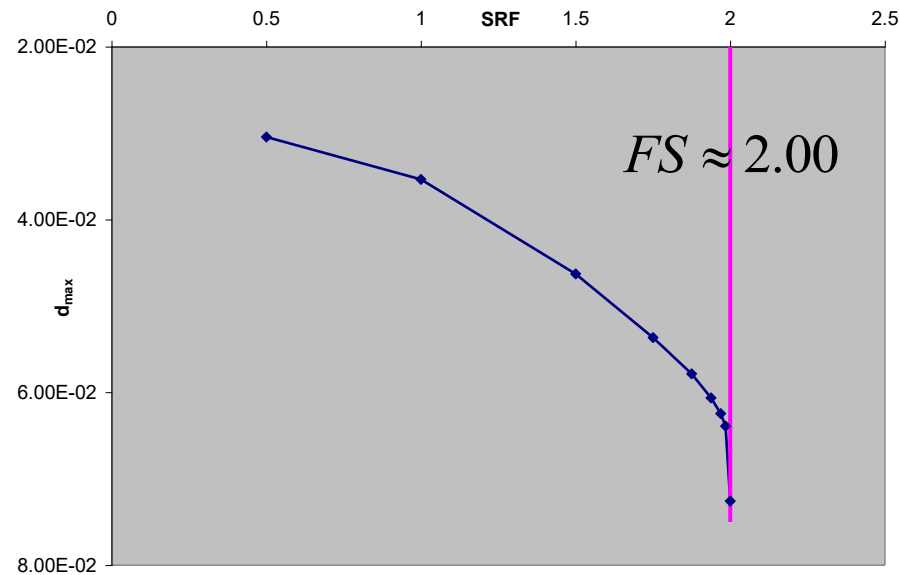


Fig 3.2 SRF vs. $d_{\max}$ for Example 3

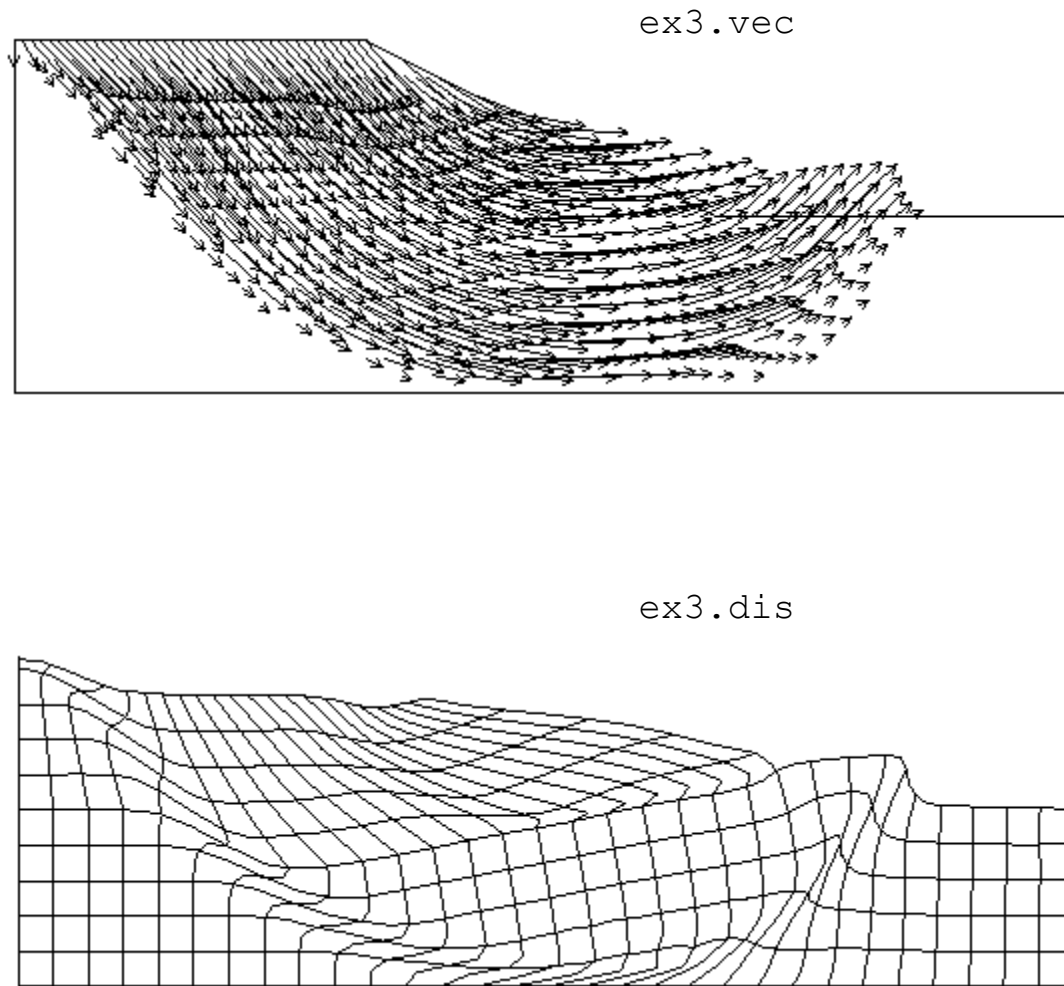


Fig 3.3 Displacement vectors and deformed mesh at failure for Example 3

Example 4: A homogeneous slope including a free-surface

The stability analysis is of the homogeneous c' - ϕ' slope including a free surface as shown in Figure 4.1. It is assumed in this case that the saturated unit weight is applicable both below and above the free-surface, so only one property type is needed.

Data for Example 4 (ex4.dat)

"Example 4: A homogeneous slope including a free-surface"

"Width of top of embankment (w1) "

25.0

"Width of sloping portion of embankment (s1) "

17.0

"Distance foundation extends to right of embankment toe (w2) "

24.0

"Height of embankment (h1) "

10.0

"Thickness of foundation layer (h2) "

10.0

"Number of x-elements in embankment (nx1) "

21

"Number of x-elements to right of embankment toe (nx2) "

12

"Number of y-elements in embankment (ny1) "

5

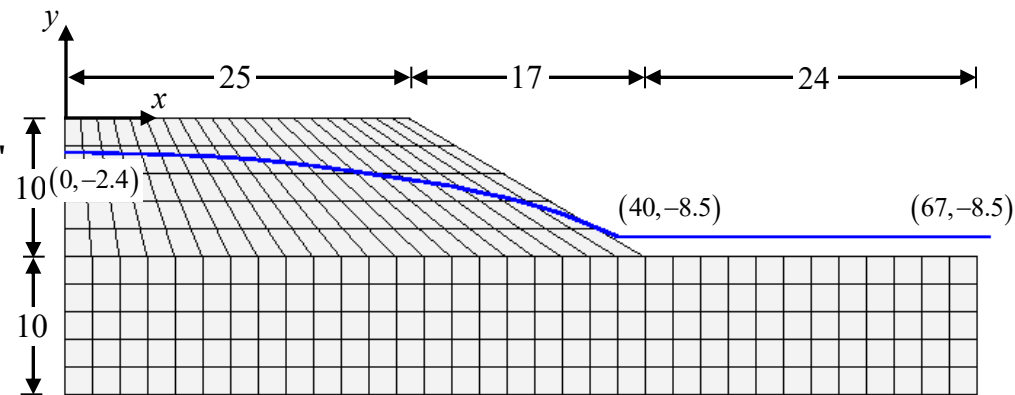


Fig 4.1 Finite element mesh for Example 4

Homogeneous
with free
surface

$\phi' = 5^\circ$
 $c' = 200$
 $\gamma = 120$

"Number of y-elements in foundation (ny2)"

5

"Number of different property groups (np_types)"

1

"Material properties (phi,c,psi,gamma,e,v) for each group"

5.0 200.0 0.0 120.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.0

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

22

0.0	-2.40	2.0	-2.45	4.0	-2.50	6.0	-2.55	8.0	-2.60
10.0	-2.67	12.0	-2.80	14.0	-2.95	16.0	-3.15	18.0	-3.40
20.0	-3.65	22.0	-3.95	24.0	-4.25	26.0	-4.55	28.0	-4.85
30.0	-5.30	32.0	-5.70	34.0	-6.25	36.0	-6.85	38.0	-7.65
40.0	-8.50	67.0	-8.50						

"Unit weight of water (gam_w)"

62.4

"Iteration ceiling (limit)"

1000

"Factor of Safety accuracy tolerance (fos_tol)"

0.05

Output for Example 4 (ex4.res)

0.5000	0.1950E+00	5
1.0000	0.2803E+00	33
1.2500	0.3630E+00	184
1.2656	0.4893E+00	1000

Estimated Factor of Safety = 1.27

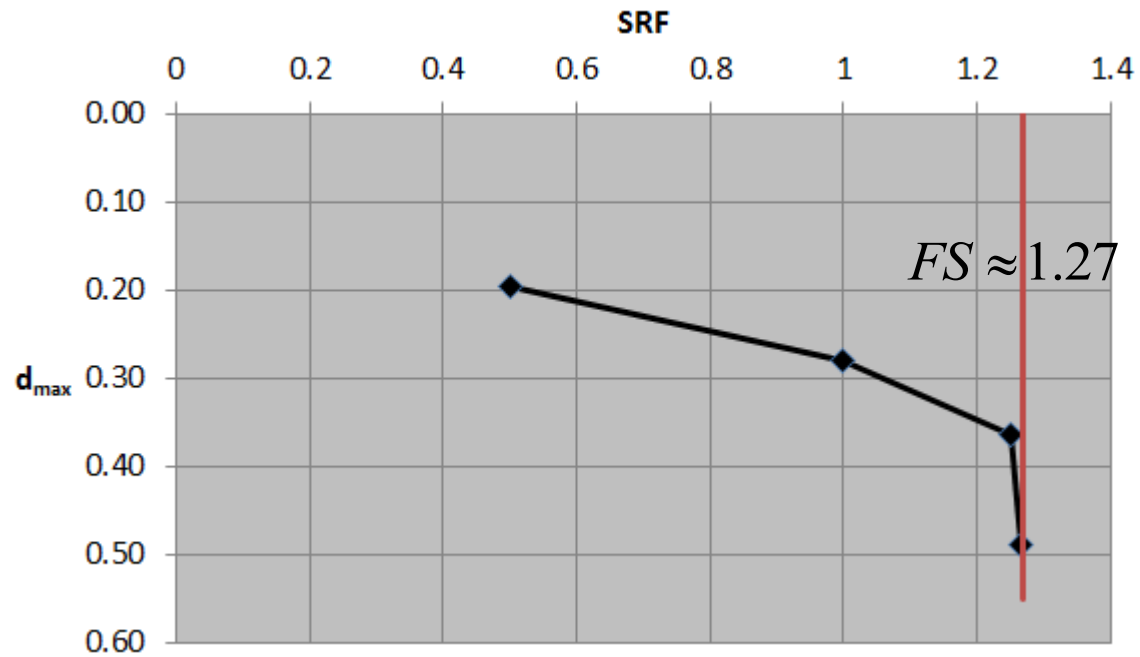


Fig 4.2 SRF vs. $d_{\max}$ for Example 4

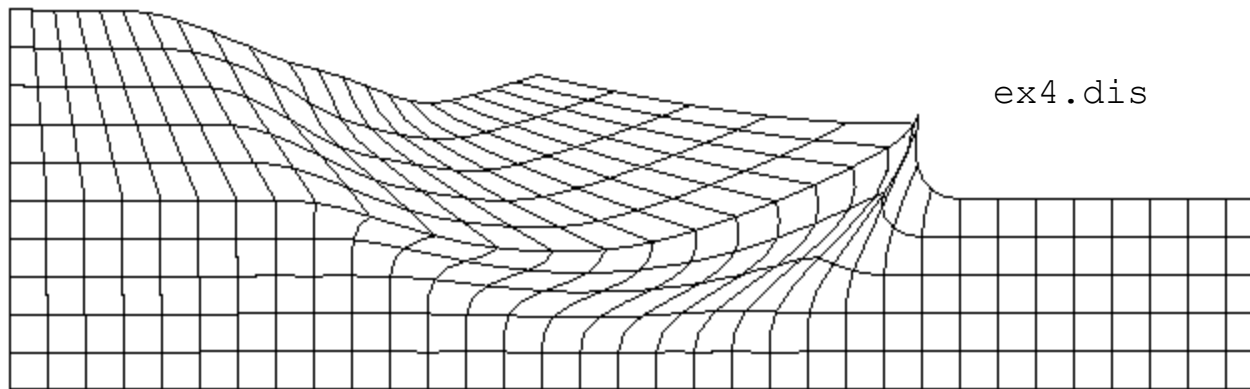
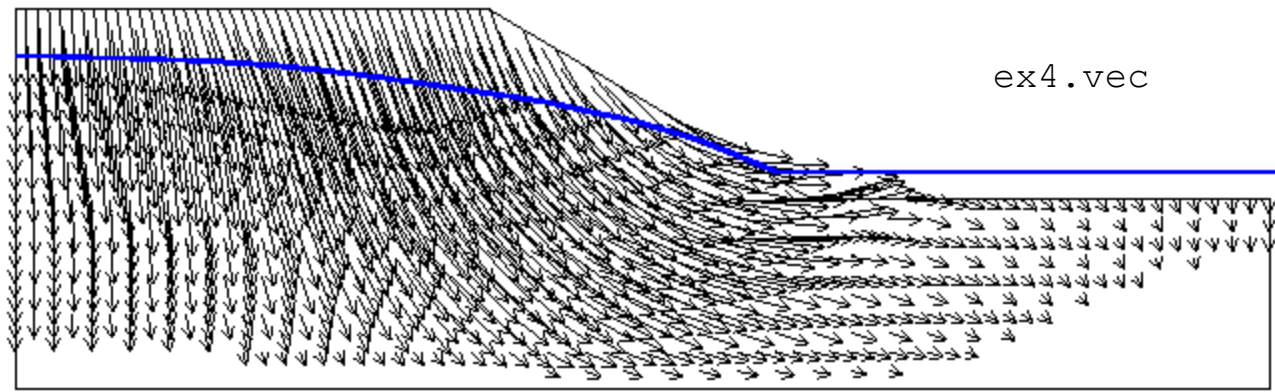


Fig 4.3 Displacement vectors and deformed mesh at failure for Example 4

Example 5: A completely submerged homogeneous slope

The stability analysis is of a homogeneous c' - ϕ' slope completely submerged beneath 2m of water as shown in Figure 5.1.

Data for Example 5 (ex5.dat)

"Example 5: A completely submerged homogeneous slope"

"Width of top of embankment (w1)"

30.0

"Width of sloping portion of embankment (s1)"

20.0

"Distance foundation extends to right of embankment toe (w2)"

0.0

"Height of embankment (h1)"

10.0

"Thickness of foundation layer (h2)"

0.0

"Number of x-elements in embankment (nx1)"

20

"Number of x-elements to right of embankment toe (nx2)"

0

"Number of y-elements in embankment (ny1)"

10

"Number of y-elements in foundation (ny2)"

0

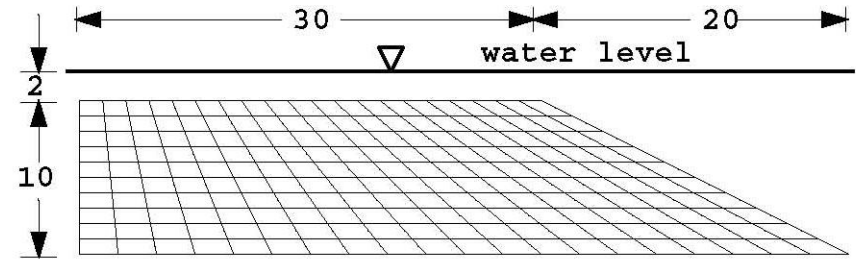


Fig 5.1 Finite element mesh for Example 5

Fully
submerged and
homogeneous

$$\phi' = 20^\circ$$

$$c' = 10$$

$$\gamma = 20$$

"Number of different property groups (np_types)"

1

"Material properties (phi,c,psi,gamma,e,v) for each group"

20.0 10.0 0.0 20.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.0

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

2

0.0 2.0

50.0 2.0

"Unit weight of water (gam_w)"

9.81

"Iteration ceiling (limit)"

500

"Factor of safety tolerance (fos_tol)"

0.05

Output for Example 5 (ex5.res)

0.5000	0.9791E-02	9
1.0000	0.1162E-01	10
1.5000	0.1285E-01	20
1.7500	0.1344E-01	36
1.8125	0.1380E-01	94
1.8281	0.1496E-01	500

Estimated Factor of Safety = 1.83

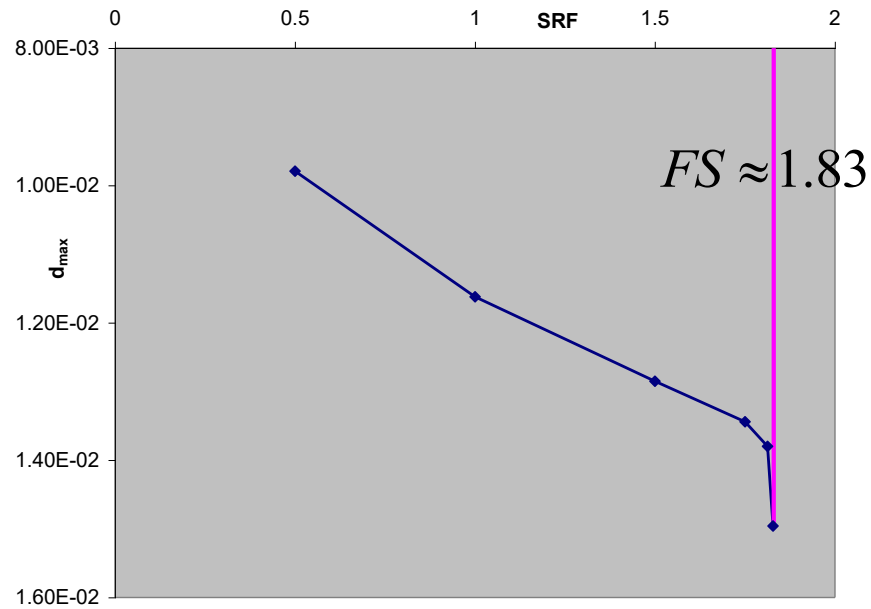


Fig 5.2 SRF vs. $d_{\max}$ for Example 5

Example 6: Rapid drawdown of a homogeneous slope from full submergence

The stability analysis is of a homogeneous $c'-\phi'$ slope (same slope as in Example 5) following 5m of rapid drawdown from full submergence as shown in Figure 6.1.

Data for Example 6 (ex6.dat)

"Example 6: Rapid drawdown of a homogeneous slope from full submergence"

"Width of top of embankment (w1)"

30.0

"Width of sloping portion of embankment (s1)"

20.0

"Distance foundation extends to right of embankment toe (w2)"

0.0

"Height of embankment (h1)"

10.0

"Thickness of foundation layer (h2)"

0.0

"Number of x-elements in embankment (nx1)"

20

"Number of x-elements to right of embankment toe (nx2)"

0

"Number of y-elements in embankment (ny1)"

10

"Number of y-elements in foundation (ny2)"

0

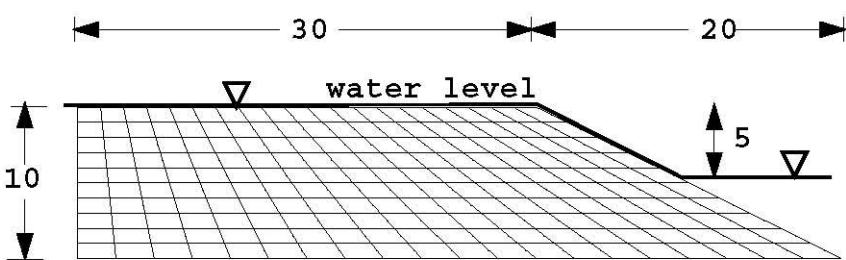


Fig 6.1 Finite element mesh for Example 6

Rapid drawdown
from full
submergence,
homogeneous

$$\phi' = 20^\circ$$
$$c' = 10$$
$$\gamma = 20$$

"Number of different property groups (np_types)"

1

"Material properties (phi,c,psi,gamma,e,v) for each group"

20.0 10.0 0.0 20.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.0

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

4

0.0 0.0

30.0 0.0

40.0 -5.0

50.0 -5.0

"Unit weight of water (gam_w)"

9.81

"Iteration ceiling (limit)"

500

"Factor of safety tolerance (fos_tol)"

0.05

Output for Example 6 (ex6.res)

0.5000	0.8548E-02	24
1.0000	0.1272E-01	111
1.0156	0.1327E-01	179
1.0312	0.1607E-01	500

Estimated Factor of Safety = 1.03

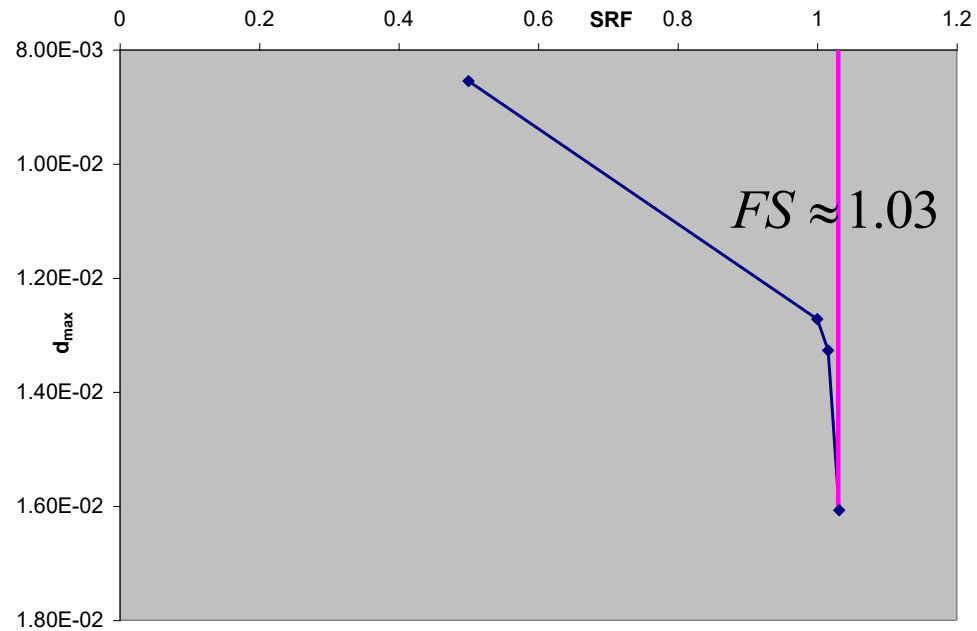
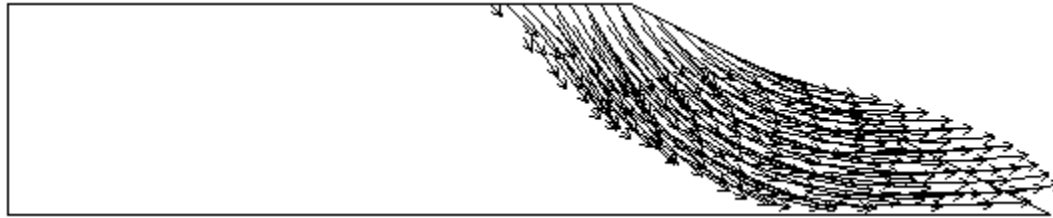


Fig 6.2 SRF vs. $d_{\max}$ for Example 6

ex6.vec



ex6.dis

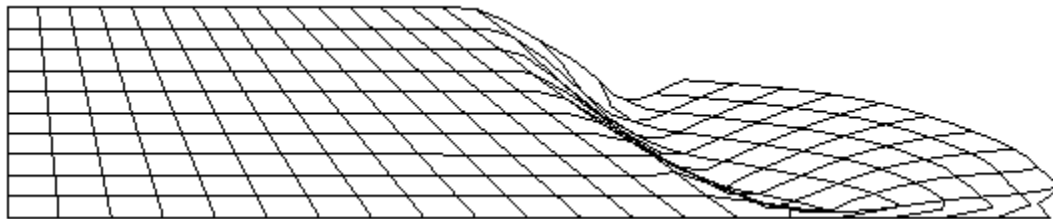


Fig 6.3 Displacement vectors and deformed mesh at failure for Example 6

Example 7: A homogeneous slope subjected to a horizontal pseudo-acceleration

The stability analysis is of a homogeneous slope subjected to a horizontal acceleration factor of $0.25g$ as shown in Figure 7.1

Data for Example 7 (ex7.dat)

'Example 7: A homogeneous slope subjected to a horizontal pseudo-acceleration'

"Width of top of embankment (w1)"
30.0

"Width of sloping portion of embankment (s1)"
20.0

"Distance foundation extends to right of embankment toe (w2)"
30.0

"Height of embankment (h1)"
10.0

"Thickness of foundation layer (h2)"
10.0

"Number of x-elements in embankment (nx1)"
50

"Number of x-elements to right of embankment toe (nx2)"
30

"Number of y-elements in embankment (ny1)"
10

"Number of y-elements in foundation (ny2)"
10

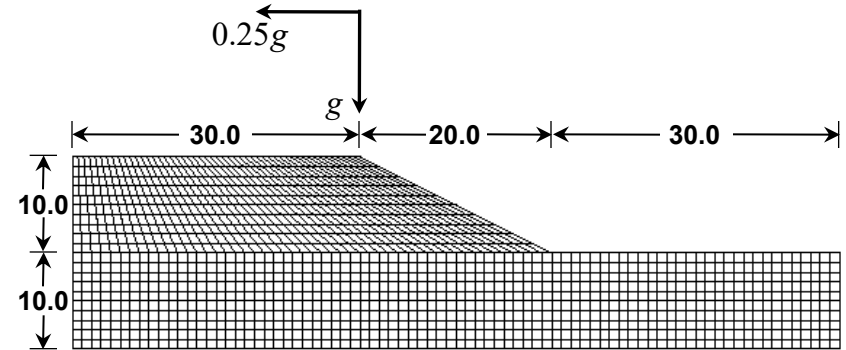


Fig 7.1 Finite element mesh for Example 7

**Homogeneous slope with
a horizontal acceleration
of $0.25g$**

$$\phi' = 30^\circ$$

$$c' = 20.0$$

$$\gamma = 20.0$$

"Number of different property groups (np_types)"

1

"Material properties (phi,c,psi,gamma,e,v) for each group"

30.0 20.0 0.0 20.0 1.e5 0.3

"Property group assigned to each element (etype, data not needed if np_types=1)"

"Pseudo-static analysis: Horizontal acceleration factor (k_h)"

0.25

"Number of free surface points and their coordinates (nosurf, surf(2,nosurf))"

0

"Unit weight of water (gam_w)"

9.81

"Iteration ceiling (limit)"

2000

"Factor of Safety accuracy tolerance (fos_tol)"

0.05

Output for Example 7 (ex7.res)

0.5000	0.4984E-01	227
1.0000	0.5313E-01	31
1.2500	0.6154E-01	79
1.3750	0.6896E-01	350
1.3906	0.6978E-01	599
1.4062	0.8632E-01	2000

Estimated Factor of Safety = 1.41

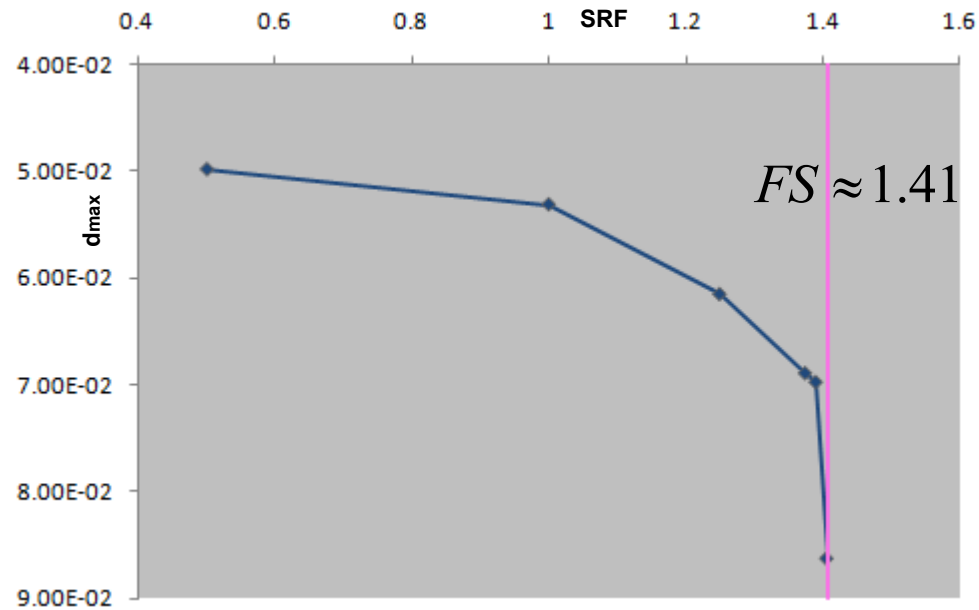


Fig 7.2 SRF vs. $d_{\max}$ for Example 7

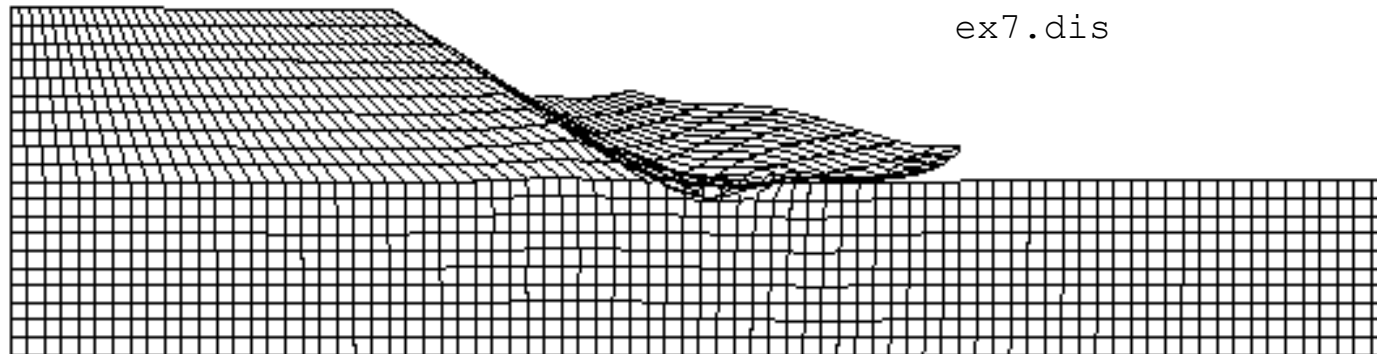


Fig 7.3 Displacement vectors and deformed mesh at failure for Example 7